Digital Twin Practice

4-Stage Supply Chain

Production Chain and Supply Policies

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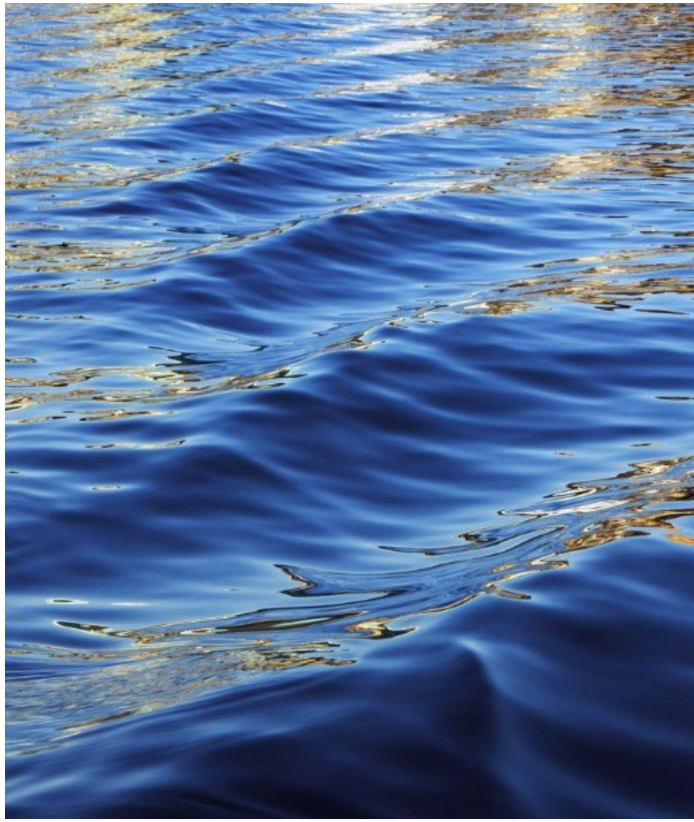

EUGLOH
European University Alliance for Global Health

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Learning objectives

1. Gain insight into the impact of production and sourcing policies on supply chain and logistics performance;
2. Develop the anyLogistix skills needed to create four-stage supply chain models, conduct experiments, and measure performance;
3. Understand the advantages and disadvantages of single versus dual sourcing strategies.

Single versus multiple suppliers

	Advantages	Disadvantages
Single Supplier	• Long-term agreements, • Price stability, • Integration in the product development process at a very early stage, • Low transactional costs, • Control of the supply chain.	• Inefficient pricing policy (no competition) • Delivery time, quality and service problems, • Lack of collaboration with many suppliers. • Major risks: dependence, CoS rupture, …
Multiple Suppliers	• Order balancing • Competition among them • Supplier risk mitigation and diversification risk mitigation and diversification breakup	• Low volume risks. • Lead-time variation • Higher management cost

Purchases from global suppliers

- For global sourcing, items with high volume, steady demand and low transportation costs are preferable. Also to be analyzed:
  - Costs: labor, taxes, transportation, insurance, transshipment, duties and transactions.
  - Quality: bill of materials, quality control, after-sales service and certifications.
  - Service: on-time delivery, responsiveness, flexibility, technical equipment, image and reliability.
  - Sustainability: political, economic and social issues.
- Advantages:
  - Access to the widest range of suppliers available.
  - Access to technology.
- Disadvantages:
  - Increased efforts to establish relationships with global suppliers or partners.
  - Currency fluctuation risk.
  - Longer supply times.
  - Stability and political interests.
  - Cultural and regulatory differences.

In this practice we will see…

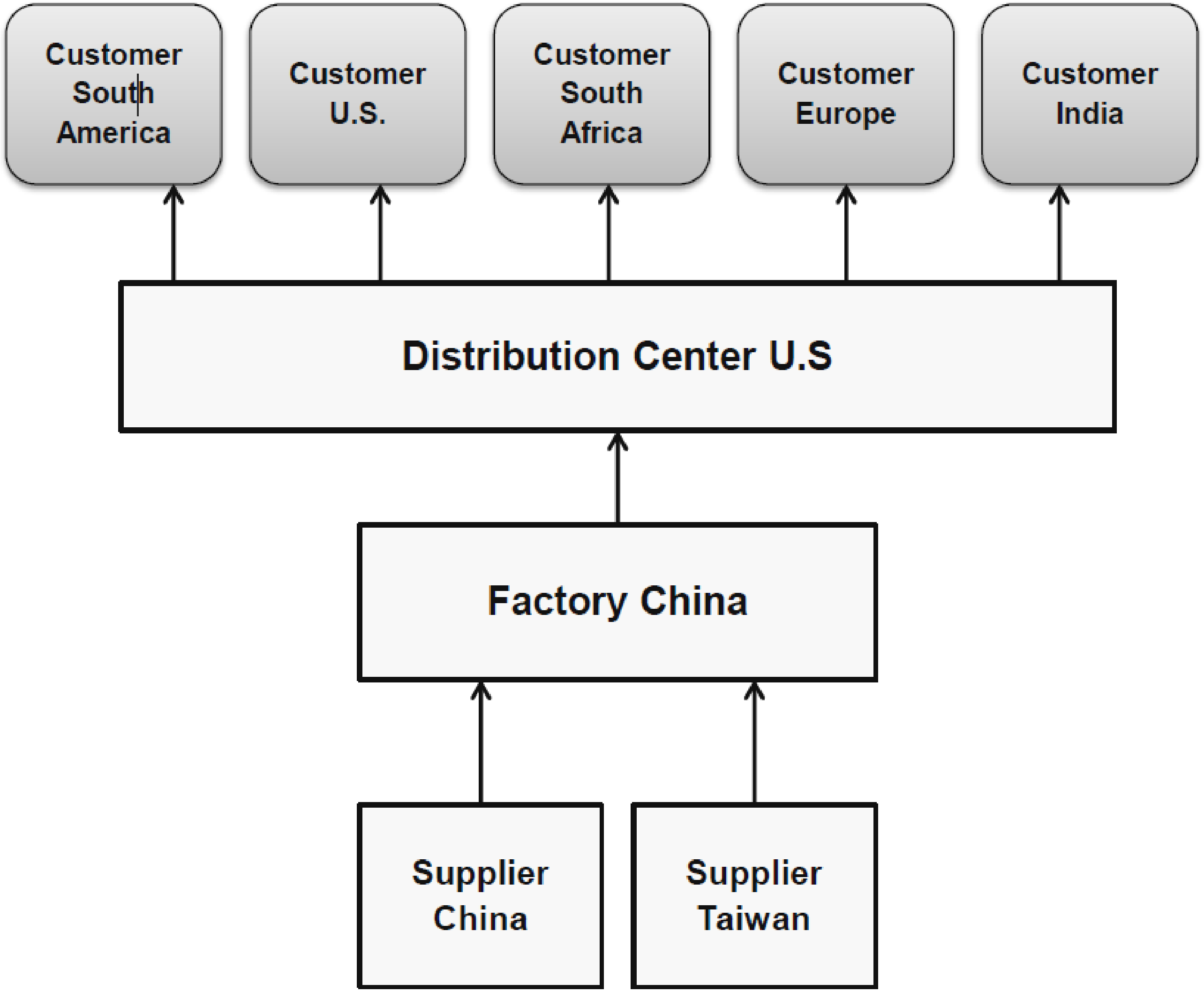
- The WHC case study.
- Exercise 1: comparison and analysis of two demand scenarios, one optimistic, the other pessimistic.
- Exercise 2: Increasing distribution center capacity and the effect of applying other inventory policies.
- Exercise 3: Dynamic Dual-Sourcing, diversification of suppliers according to demand.
- Exercise 4: Static Dual-Sourcing, pre-allocation of distribution centers.

Case Study: WHC mobile vendor

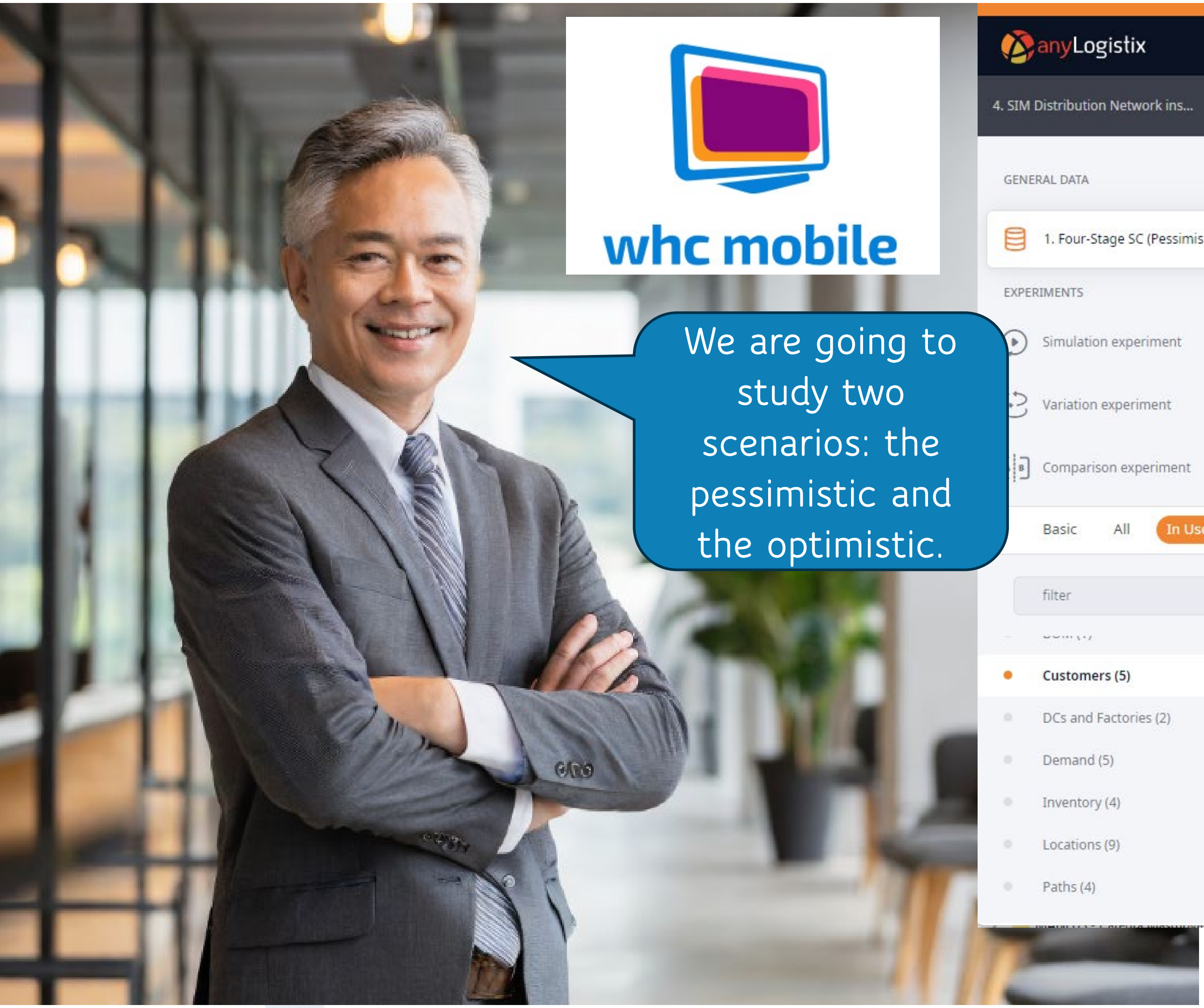
WHC Mobile

whc mobile

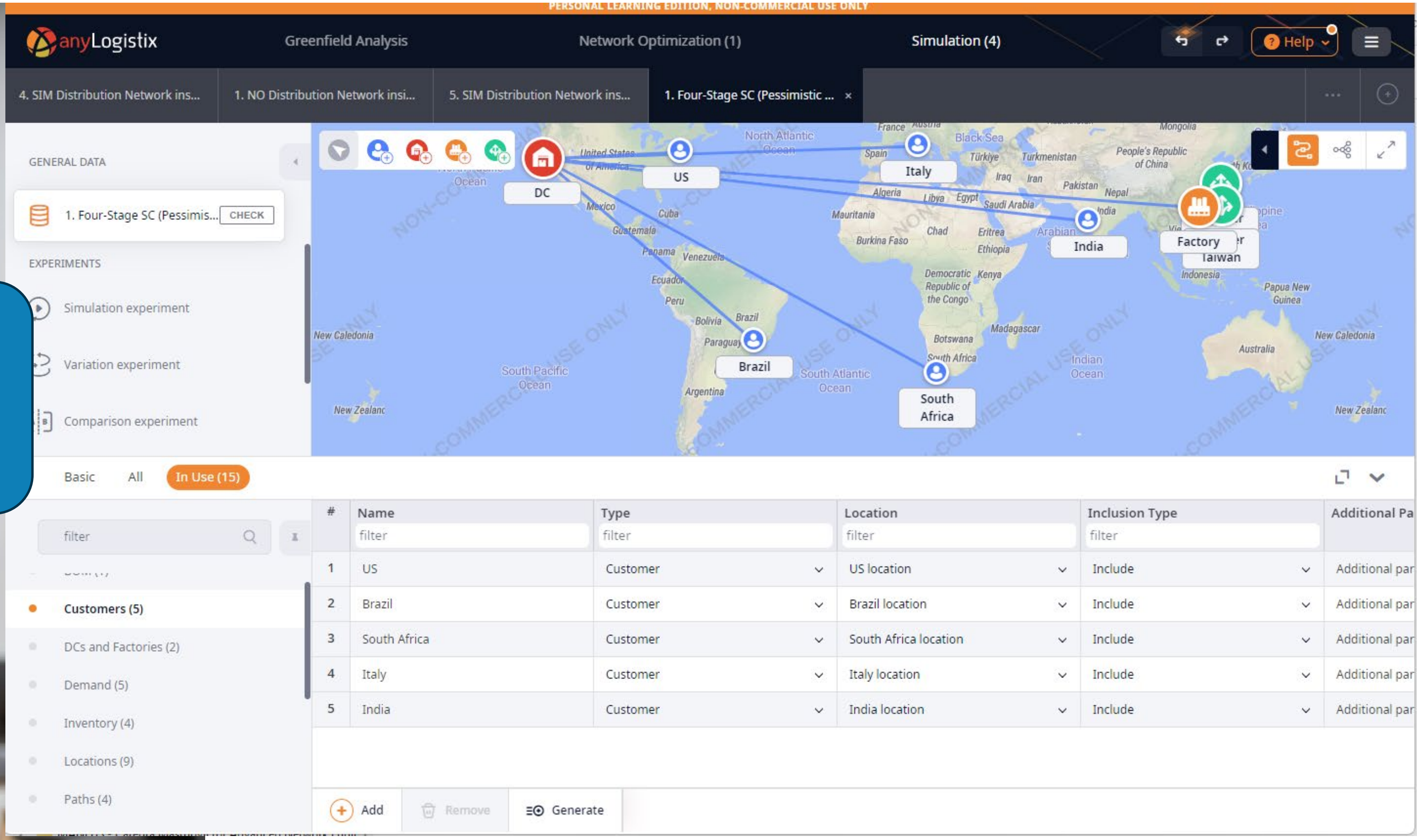
- Headquartered in Hong Kong.
- Supply chain for the production and distribution of smartphones.
- Production in China (assembly).
  - The factory in China is dedicated to assembly.
- Two main components:
  - One screen, manufactured in China and delivered by truck.
  - Two chips, manufactured in Taiwan, and received via ferry.
- The factory delivers the smartphones by air to the d center in the US.
- From the USA, the distribution center sends them by customers around the world (diagram).
- The factory and distribution center are running a Min-Max inventory control policy at a 1% interest rate.



WHC Supply Chain



We are going to study two scenarios: the pessimistic and the optimistic.



Scenarios 1, 2: optimistic and pesimistic demand situations

As always, the scenario must be configured

- Products:
  - Those we consume and those we produce

#	Name	Unit	Selling Price	Cost	Cost Unit
1	Smartphone	pcs	600	200	USD
2	Display	pcs	30	10	USD
3	Chip	pcs	20	5	USD

- Conversion of product units:

#	Product	Amount from	Unit from		Amount to	Unit to
1	Smartphone	1	pcs	=	0.001	m³
2	Display	1	pcs	=	0.0005	m³
3	Chip	1	pcs	=	0.000001	m³

Production configuration

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- BOM - Bill of Materials:
  - As we are in a PRODUCTION scenario, we have to indicate the relationship between the different materials: "to produce a smartphone, you need a screen and two chips".

#	Name	End Product	Quantity	Co-products	Components
	Filter	Filter	Filter	Filter	Filter
1	BOM 1	Smartphone	1	[]	[Display:1.0, Chip:2.0]

- Production policy:
  - What we produce, from where, at what cost.

#	Site	Product	Type	Parameters	BOM	Production Cost	Currency	CO2 per product	Time F
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	Factory	Smartphone	Simple make pol.	Time = 0.01 (day)	BOM 1	50	USD	0	(All pe

As always, the scenario must be configured

- Vehicles (means of transportation):

#	Name	Capacity	Capacity Unit	Speed	Speed Unit
1	Airplane	40	m^3	800.0	km/h
2	Truck	20	m^3	50.0	km/h
3	Ship	2,000	m^3	50.0	km/h
4	Ferry	2,000	m^3	50.0	km/h

As always, the scenario must be configured

- Sourcing policies for each node:

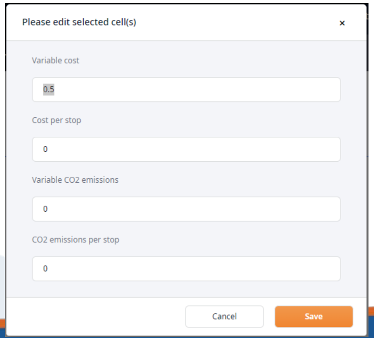
#	Delivery Destinat...	Product	Type	Parameters	Sources	Time Period	Inclusion Type
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	Factory	Display	Closest (Fixed So.	No parameters	Supplier China	(All periods)	Include
2	Factory	Chip	Closest (Fixed So.	No parameters	Supplier Taiwan	(All periods)	Include
3	DC	Smartphone	Closest (Fixed So.	No parameters	Factory	(All periods)	Include
4	(All customers)	Smartphone	Closest (Fixed So.	No parameters	DC	(All periods)	Include

As always, the scenario must be configured

- Transportation routes:

#	From	To	Cost Calculation	Cost Calculation ...	...	Curre...	...	Distance ...	Transp...	Time ...	Straight	Vehicle ...	Transport
	Filter	Filter	Filter	Filter		Filter		Filter	Filter	Filter	Filter	Filter	Filter
1	Supplier China	Factory	Distance-based	0.5 * distance + 0	...	USD	0	km	0	day		Truck	LTL
2	Supplier Taiwan	Factory	Distance-based	0.8 * distance + 0	...	USD	0	km	0	day		Ferry	LTL
3	Factory	DC	Product&distanc...	0.01 * product (m...	...	USD	0	km	2	day		Airplan	LTL
4	DC	(All loca.	Product&distanc...	0.01 * product (m...	...	USD	0	km	2	day		Airplan	LTL

- Land and sea routes: based on cost per km and cost per stop.
- Air routes: based on product and distance.
- Each one with its specific parameterization.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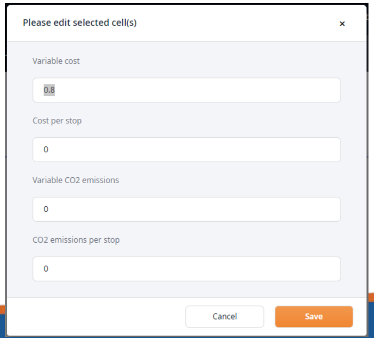
Variable cost: 0.5
Cost per stop: 0
Variable CO2 emissions: 0
CO2 emissions per stop: 0

Cancel | Save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Variable cost: 0.8
Cost per stop: 0
Variable CO2 emissions: 0
CO2 emissions per stop: 0

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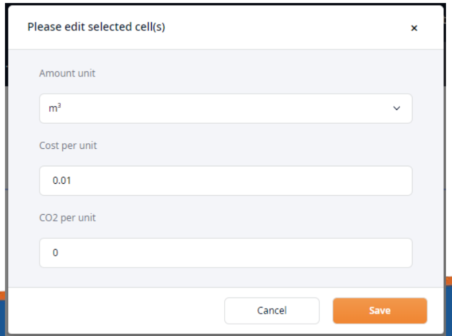
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Amount unit: m^3
Cost per unit: 0.01
CO2 per unit: 0

Cancel | Save

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As always, the scenario must be configured

whc mobile

- Inventory control policies:

#	Facility	Product	Policy Type	Policy Parameters	Initial Stock, units	Periodic Check
	Filter	Filter	Filter	Filter	Filter	Filter
1	DC	Smartphone	Min-max policy	s=20, S=50	40	
2	Factory	Smartphone	Min-max policy	s=30, S=60	40	
3	Factory	Chip	Unlimited inventory	Unlimited	N/A	
4	Factory	Display	Unlimited inventory	Unlimited	N/A	

Positive and negative demand scenarios

- DEMAND configuration in a pessimistic scenario:

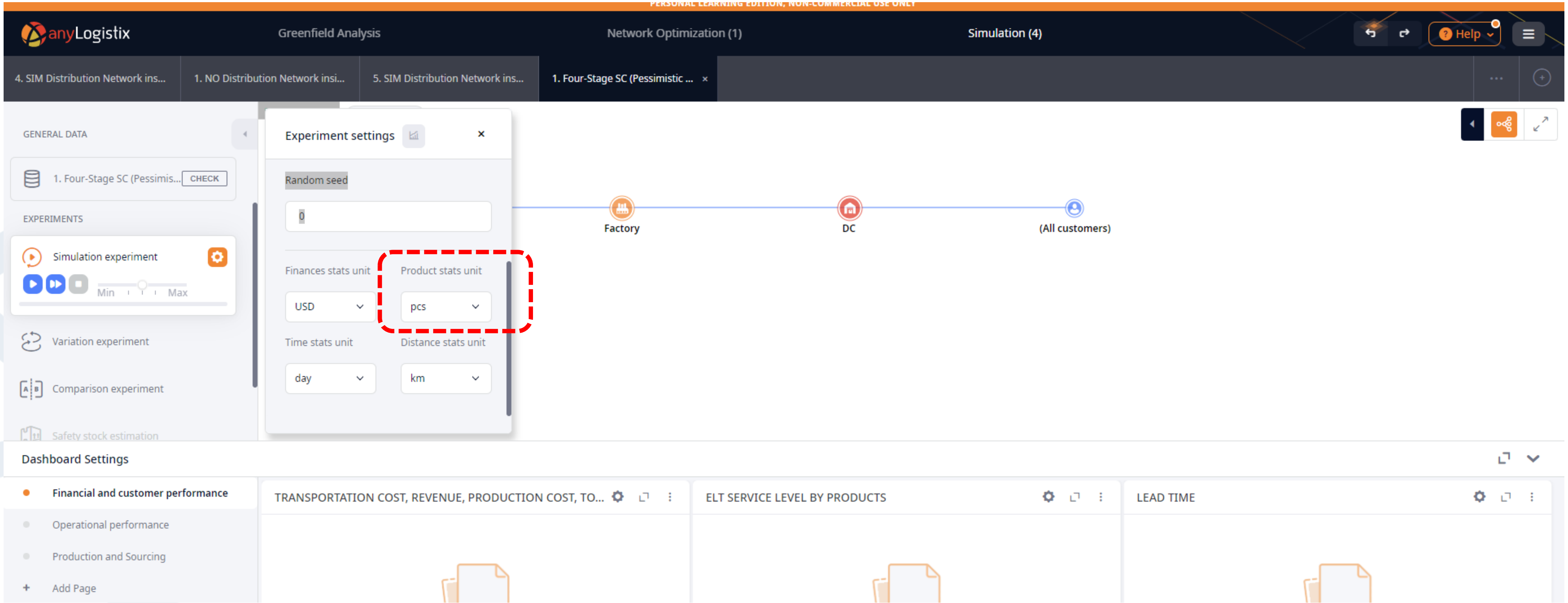
#	Customer	Product	Demand Type	Parameters	Time Period	Rev...	Currency	Expec...	Time Unit
1	US	Smartphone	Periodic demand	Order interval=10, Quantity=7, First occurrence: First day	(All periods)	0	USD	30	day
2	Brazil	Smartphone	Periodic demand	Order interval=10, Quantity=3, First occurrence: First day	(All periods)	0	USD	30	day
3	South Africa	Smartphone	Periodic demand	Order interval=10, Quantity=2, First occurrence: First day	(All periods)	0	USD	30	day
4	Italy	Smartphone	Periodic demand	Order interval=10, Quantity=2, First occurrence: First day	(All periods)	0	USD	30	day
5	India	Smartphone	Periodic demand	Order interval=10, Quantity=6, First occurrence: First day	(All periods)	0	USD	30	day

- DEMAND configuration in an optimistic scenario:

#	Customer	Product	Demand Type	Parameters	Time Period	Rev...	Currency	Expec...	Time Unit
1	US	Smartphone	Periodic demand	Order interval=10, Quantity=35, First occurrence: First day	(All periods)	0	USD	30	day
2	Brazil	Smartphone	Periodic demand	Order interval=10, Quantity=15, First occurrence: First day	(All periods)	0	USD	30	day
3	South Africa	Smartphone	Periodic demand	Order interval=10, Quantity=10, First occurrence: First day	(All periods)	0	USD	30	day
4	Italy	Smartphone	Periodic demand	Order interval=10, Quantity=10, First occurrence: First day	(All periods)	0	USD	30	day
5	India	Smartphone	Periodic demand	Order interval=10, Quantity=30, First occurrence: First day	(All periods)	0	USD	30	day

We prepare the simulation

- The product unit with which we will simulate now is "units" instead of m3, since we have to manufacture.



We prepare the scorecard to be obtained

- Financial and customer performance:
  - Cost of production, Profit, Revenue, Total cost, Transportation cost (chart)
  - ELT service level per order (chart)
  - Lead-time (graph)
- Operational performance:
  - Maximum capacity (graph)
  - Available inventory (graph)
- Production and supply:
  - Production cost, transportation cost (table, per item)
  - Demand (backorders)
  - Demand realized (abandoned orders) per customer,
  - Realized demand (orders) per customer
  - Fulfillment (backorders)
  - Fulfillment received (orders on time)
  - Fulfillment received (orders) by customer
  - Products produced

Ex.1/2 Simulation of SC status neg/pos

- Financial scorecard (PESSIMISTIC):

Financial and customer performance
Operational performance
Production and Sourcing
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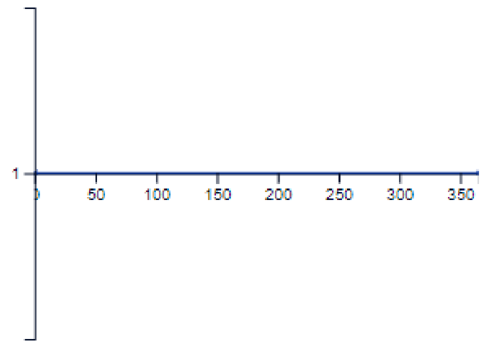
TRANSPORTATION COST, REVENUE, ...

Statistics	Value	Unit
filter	filter	filter
Profit	405,346.393	USD
Production Cost	38,500	USD
Revenue	444,000	USD
Total Cost	38,653.607	USD
Transportation Cost	153.607	USD

ELT SERVICE LEVEL BY PRODUCTS



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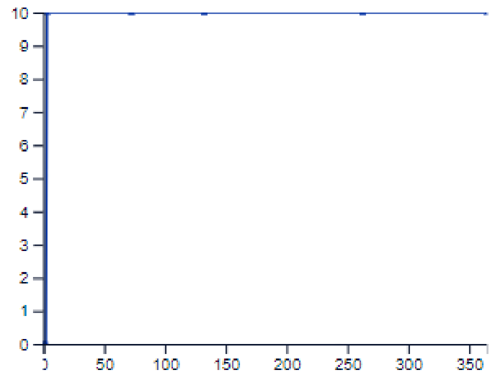

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- Financial scorecard (OPTIMIST):

Financial and customer performance
Operational performance
Production and Sourcing
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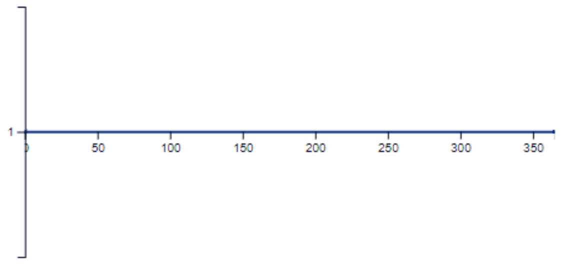
TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,007,437.706	USD
2	Production Cost	93,250	USD
3	Revenue	1,101,000	USD
4	Total Cost	93,562.294	USD
5	Transportation Cost	312.294	USD

ELT SERVICE LEVEL BY PRODUCTS



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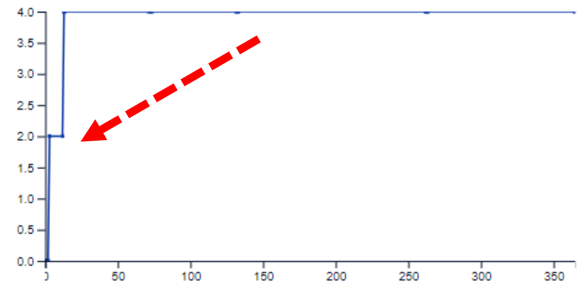

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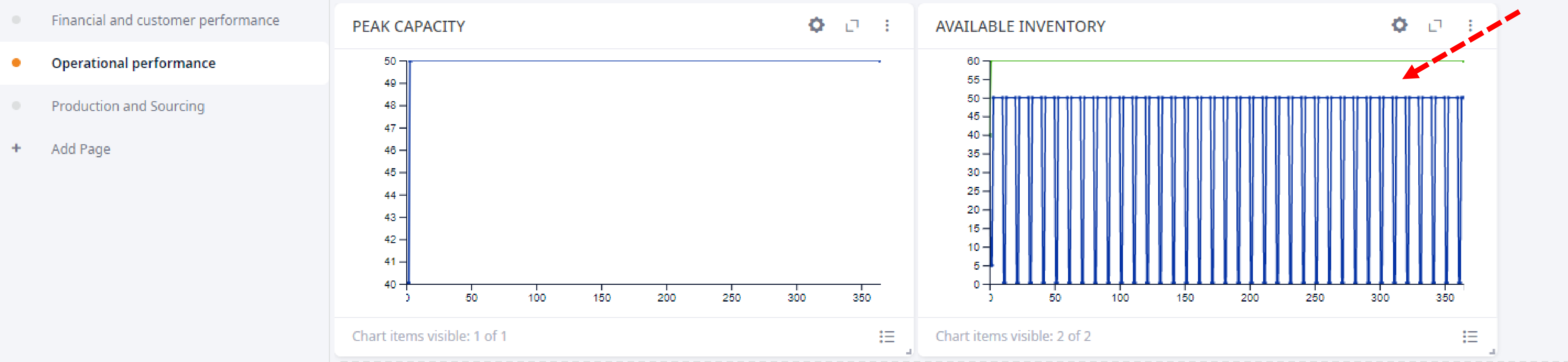
Ex.1/2 Simulation of SC status neg/pos

- Operating efficiency chart (PESSIMISTIC):

- Operational efficiency chart (OPTIMIST):



Ex.1/2 Simulation of SC status neg/pos

- Production and supply scorecard (PESIMISTIC):

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	38,500	USD
2	Transportation Cost	DC	71.552	USD
3	Transportation Cost	Factory	82.054	USD

DEMAND (PRODUCTS BACKLOG), PRODUCTS PRODUCED, FULFILLMENT (LATE PRODUCTS), FULFILL...

#	Statistics	Product	Value	Unit
1	Demand (Products Backlog)	Chip	0	pcs
2	Demand (Products Backlog)	Display	0	pcs
3	Demand (Products Backlog)	Smartphone	0	pcs
4	Fulfillment Received (Products) by Customer	Smartphone	740	pcs
5	Fulfillment Received (Products On-time)	Smartphone	740	pcs
6	Products Produced	Smartphone	770	pcs

- Production and supply scorecard (OPTIMIST):

Financial and customer performance
Operational performance
Production and Sourcing
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TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	93,250	USD
2	Transportation Cost	DC	110.44	USD
3	Transportation Cost	Factory	201.854	USD

DEMAND (PRODUCTS BACKLOG), PRODUCTS PRODUCED, F...

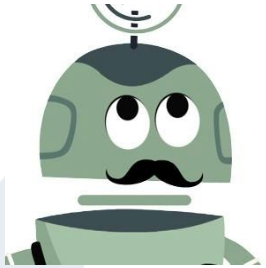
#	Statistics	Product	Value	Unit
1	Demand (Products B...	Chip	0	pcs
2	Demand (Products B...	Display	0	pcs
3	Demand (Products B...	Smartphone	0	pcs
4	Fulfillment Received ...	Smartphone	1,835	pcs
5	Fulfillment Received ...	Smartphone	1,835	pcs
6	Products Produced	Smartphone	1,865	pcs

Ex.1/2 Scenario comparison

Table 15: KPI comparison.

KPI	Pessimistic Scenario	Optimistic Scenario
Financial and customer performance:		
Production cost, $	38 500.0	93 250.0
Profit, $	405 347.393	1 007 437.706
Revenue, $	444 000.0	1 101 000.0
Total cost, $	38 653.607	93 562.294
Transportation cost (distribution center), $	71.552	110.44
Transportation cost (Factory), $	82.054	201.854
Service level, %	100%	100%
Lead time, days	10	4
Operational performance:		
Maximum capacity usage in the supply chain, pcs	50	50
Maximum inventory in the supply chain (distribution center), pcs	50	50
Maximum inventory in the supply chain (Factory), pcs	60	60
Production and sourcing performance:		
Current backlog orders	0	0
Customer delayed orders	0	0
Customer dropped orders	0	112.0
Customer in-time orders	185.0	73.0
Customer orders	185.0	185.0
Customer orders arrived	185.0	73.0
Produced, pcs	730.0	1865.0

Ex.1/2 Scenario comparison



- The pessimistic and optimistic scenarios are compared in several key respects in the document provided:

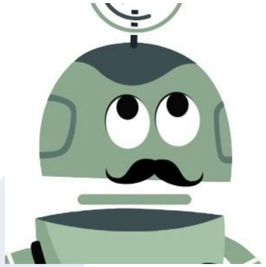
1. Financial and customer performance:
    1. **Cost of production**: In the pessimistic scenario, the cost of production is 38,500.0, *while in the optimistic scenario it is*93,250.0.
    2. **Benefit**: The benefit in the pessimistic scenario is 405,347,393, *and in the optimistic scenario it is 1*,007,437,706.
    3. **Revenues**: Revenues are 444,000.0 *in the pessimistic scenario and 1*,101,000.0 in the optimistic scenario.
    4. **Total cost**: In the pessimistic scenario, the total cost is 38,653,607, *and in the optimistic scenario it is* 93,562,294.
    5. **Transportation cost**: Both the transportation cost at the distribution center and at the factory is higher in the optimistic scenario compared to the pessimistic scenario.
    6. **Level of service**: The level of service is 100% in both scenarios.
    7. **Delivery time**: Delivery time is 10 days in the pessimistic scenario and 4 days in the optimistic scenario.

- These data clearly show significant differences in financial and customer performance between the pessimistic and optimistic scenarios, highlighting the importance of considering different conditions and demands in supply chain management

2. Production and supply capacity and performance:
    2. In the optimistic scenario, there is an increase in production capacity and in the quantity of products produced compared to the pessimistic scenario.
    3. It is mentioned that the increase in demand has led to a higher profit in the supply chain, but also to a decrease in order fulfillment rates. This highlights the need to redesign the supply chain to adapt to optimistic scenarios

- In summary, the comparison between the pessimistic and optimistic scenario highlights the significant implications that different conditions can have on financial performance, production capacity and customer satisfaction in the supply chain. These findings can be critical to making informed decisions and improving operational efficiency in variable and changing environments.

Ex.1/2 Scenario comparison

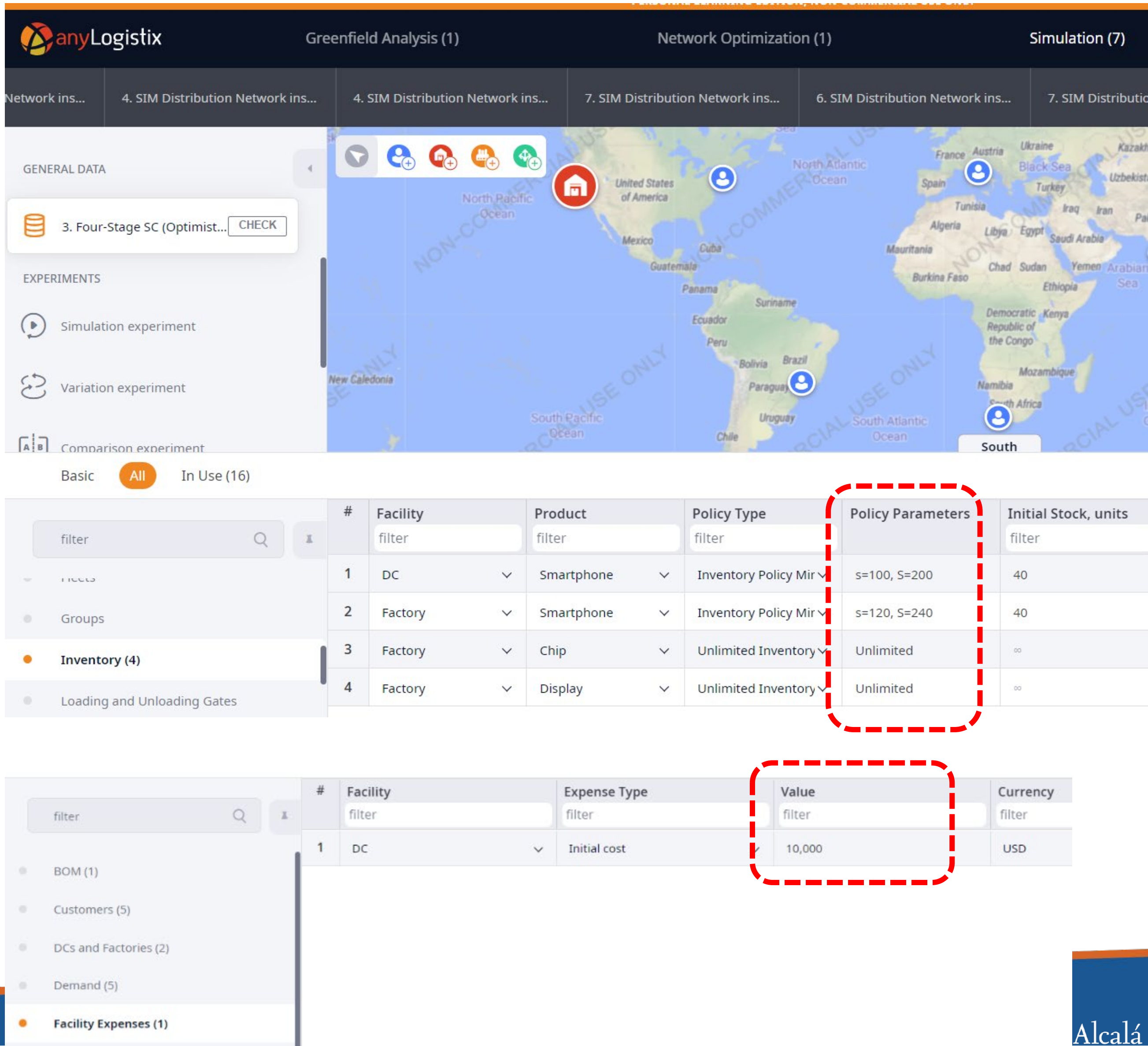


- **Reasons for the differences:**
  - The differences in results between the pessimistic and optimistic scenarios can be attributed to several factors, such as market demand, supply chain capacity, and production and supply performance. In the optimistic scenario, higher profit and demand are observed, indicating a favorable market. However, it is also mentioned that order fulfillment rates have decreased, suggesting that the current supply chain may have limitations in terms of capacity and design. This indicates the need to redesign the supply chain to adapt to realistic optimistic scenarios and improve operational and financial performance.
- Below are the pros and cons of two options based on the information provided in the document:
- **Option 1: Increase the capacity of the distribution center and apply new MinMax values in inventory control.**
  - Pros:
    - Increased capacity in the distribution center to meet demand.
    - Ability to maintain a larger inventory both in the distribution center and in the factory.
  - Increased likelihood of fulfilling customer orders.
  - Cons:
    - Fixed cost of $10,000 to increase distribution center capacity.
    - Possible increase in transportation costs due to higher inventory.
- **Option 2: Build a second distribution center in China and implement the dual sourcing strategy.**
  - Pros:
    - Increased distribution capacity by having two centers in strategic locations.
    - Reduced delivery time by having an additional distribution center in China.
    - Greater flexibility in supply chain management by having two sources of supply.
  - Cons:
    - Significantly higher cost of $50,000 to build a new distribution center.
    - Possible increase in transportation costs due to the need to ship products to two different locations.

Scenario 3: Increase our inventory to improve efficiency

Increasing capacity and modifying inventory policy

- The first of the options for improving the situation is to increase the capacity of the distribution center:
  - (Optimistic scenario)
  - Apply new inventory policy values:
    - Min-Max 100-200 at the distribution center.
    - Min-Max 120-240 at factory
  - Fixed costs increase to: $10,000



Run the simulation and compare

- **Financial** efficiency before:

- Financial and customer performance
- Operational performance
- Production and Sourcing
- + Add Page

TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,007,437.706	USD
2	Production Cost	93,250	USD
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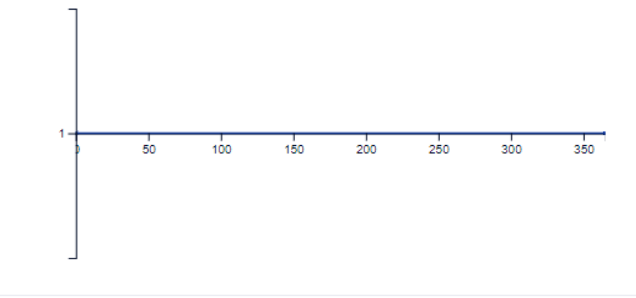
ELT SERVICE LEVEL BY PRODUCTS



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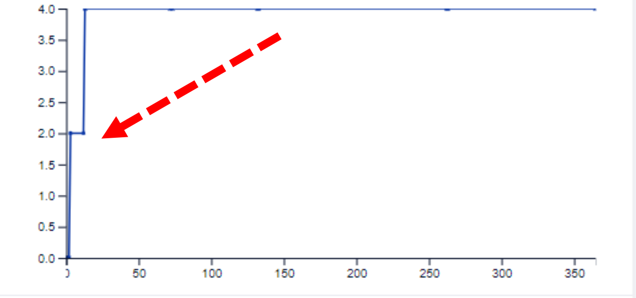

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- **Financial** efficiency after the changes:

- Financial and customer performance
- Operational performance
- Production and Sourcing
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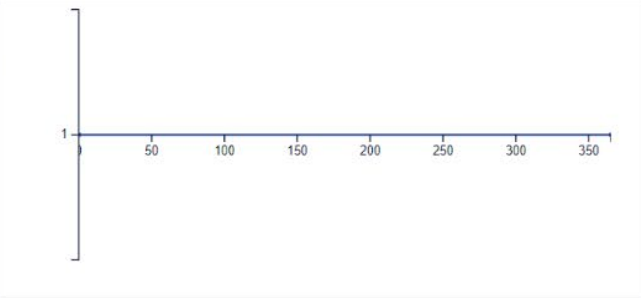
TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,985,485.255	USD
2	Production Cost	194,750	USD
3	Revenue	2,181,000	USD
4	Total Cost	195,514.745	USD
5	Transportation Cost	764.745	USD

ELT SERVICE LEVEL BY ORDERS



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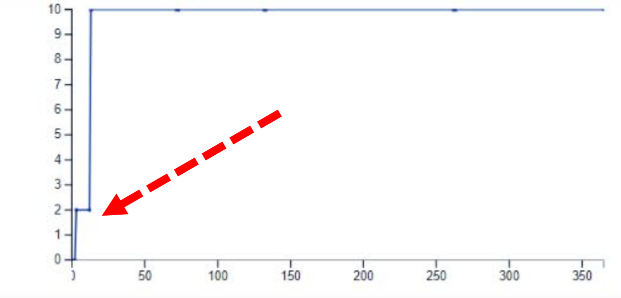

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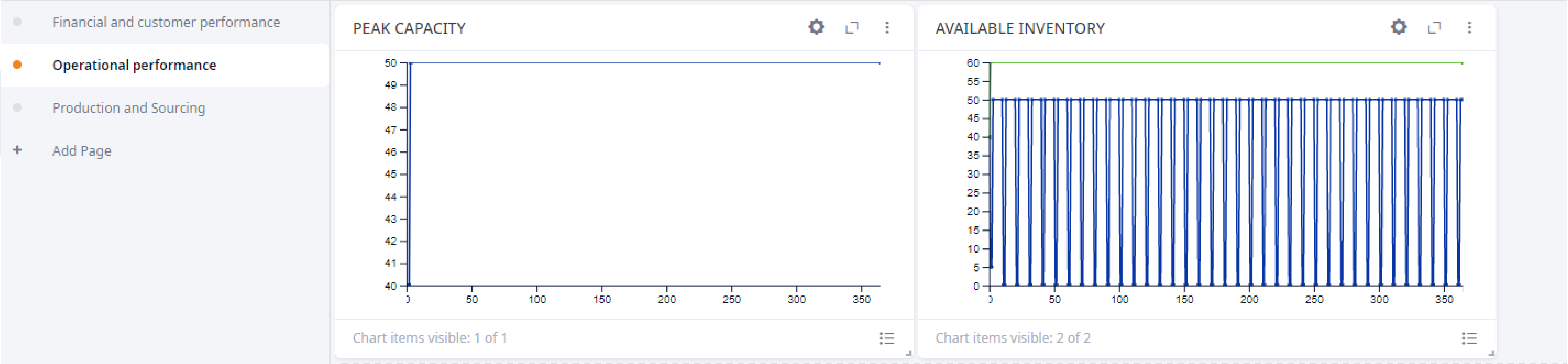
Run the simulation and compare

- Operational efficiency before:



- Operational efficiency after changes:

Run the simulation and compare

- Efficient **production and supply** before:

Financial and customer performance	
Operational performance	
Production and Sourcing	
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TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
	filter	filter	filter	filter
1	Production Cost	Factory	93,250	USD
2	Transportation Cost	DC	110.44	USD
3	Transportation Cost	Factory	201.854	USD

DEMAND (PRODUCTS BACKLOG), PRODUCTS PRODUCED, F...

#	Statistics	Product	Value	Unit
	filter	filter	filter	filter
1	Demand (Products B...	Chip	0	pcs
2	Demand (Products B...	Display	0	pcs
3	Demand (Products B...	Smartphone	0	pcs
4	Fulfillment Received ...	Smartphone	1,835	pcs
5	Fulfillment Received ...	Smartphone	1,835	pcs
6	Products Produced	Smartphone	1,865	pcs

- **Production and supply** efficiency after the changes:

Financial and customer performance	
Operational performance	
Production and Sourcing	
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TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
	filter	filter	filter	filter
1	Production Cost	Factory	194,750	USD
2	Transportation Cost	DC	349.55	USD
3	Transportation Cost	Factory	415.196	USD

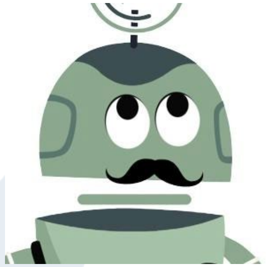
DEMAND (ORDERS BACKLOG), PRODUCTS PRODUCED, FULF...

#	Statistics	Value	Unit
	filter	filter	filter
1	Demand (Orders Back...	0	Order
2	Demand Placed (Drop...	4	Order
3	Demand Placed (Orde...	185	Order
4	Fulfillment Received (...	181	Order
5	Fulfillment Received (...	181	Order
6	Products Produced	3,895	pcs

Scenario comparison

KPI	Optimistic Scenario AS-IS Supply Chain Design	Optimistic Scenario Redesign “single distribution center - increased capacity”
Financial and customer performance:		
Production cost, $	93 250.0	194 000.0
Profit, $	1 007 437.706	1 985 485.255
Revenue, $	1 101 000.0	2 181 000.0
Total cost, $	93 562.294	195 514.745
Transportation cost (distribution center), $	110.44	764.745
Transportation cost (Factory), $	201.854	411.37
Service level, %	100%	100%
Lead time, days	4	10
Operational performance:		
Maximum capacity usage in the supply chain, pcs	50	200
Maximum inventory in the supply chain (distribution center), pcs	50	200
Maximum inventory in the supply chain (Factory), pcs	60	240
Production and sourcing performance:		
Current backlog orders	0	0
Customer delayed orders	0	0
Customer dropped orders	112.0	4.0
Customer in-time orders	73.0	181.0
Customer orders	185.0	185.0
Customer orders arrived	73.0	181.0
Produced, pcs	1865.0	3 895.0

Scenario comparison



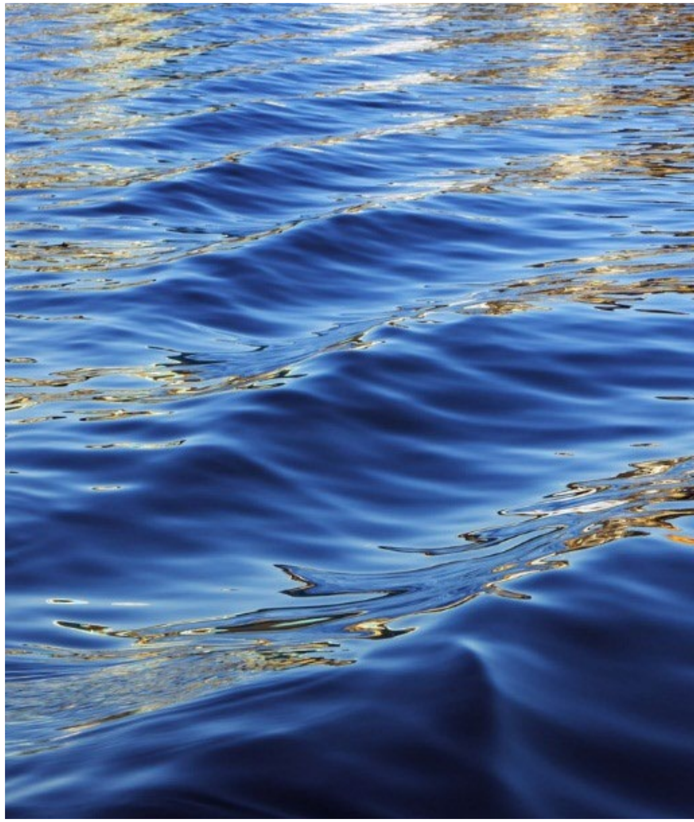
- To compare the models in the table and highlight the pros and cons of each option, as well as justify and analyze the reasons for the change, we can rely on the data provided on pages 116 and 117 of the document.
- **Optimistic Scenario AS-IS:**
  - Pros:
    - Lower transportation cost from the distribution center (110.44) compared to the factory (201.854) .
    - Shorter delivery time of 4 days compared to 10 days in the optimistic scenario.
    - Maximum capacity for use in the supply chain of 50 units .
    - Maximum inventory capacity in the distribution center of 50 units and in the factory of 60 units.
  - Cons:
    - Higher transportation cost from the factory.
    - Longer lead time compared to the optimistic scenario.
    - Lower capacity utilization and inventory in the supply chain compared to the optimistic scenario.
- **Redesign of the "Single Distribution Center - Increased Capacity" (TO-BE) Scenario:**
  - Pros:
    - Significant reduction in production cost (93,250) compared to the AS-IS design (194,000).
    - Substantial increase in profit (1,007,437,706) compared to the AS-IS design (1,985,485,255).
    - Increased production and inventory capacity in the supply chain.
    - Improved operational and financial performance.
  - Cons:
    - An initial investment may be required to establish the new distribution center in China.
    - Change in sourcing policy that may involve adjustments in supply chain management.
    - Justification and Analysis of Change:
- **Optimistic Scenario AS-IS:**
  - The shift to a new supply chain design with a single distribution center and increased capacity is justified by significant benefits in terms of cost, profit and operational performance.
  - The reduction in production costs and the increase in profit indicate a substantial improvement in the efficiency and profitability of the system.
  - The need to adapt to market demands and improve competitiveness may have been a key factor in the decision to make this change.
  - The implementation of a new distribution center in China and the adoption of a dual sourcing policy can be strategies to optimize the supply chain and meet customer needs more effectively.
  - In summary, the shift to a new supply chain design with a single distribution center and increased capacity appears to be a sound decision based on the demonstrated benefits in terms of financial and operational performance.

Scenario 4: Implement Dual-Sourcing to increase efficiency (and resilience)

EUGLOH

European University Alliance for Global Health

https://youtu.be/XJGwxb7ro-U?t=84

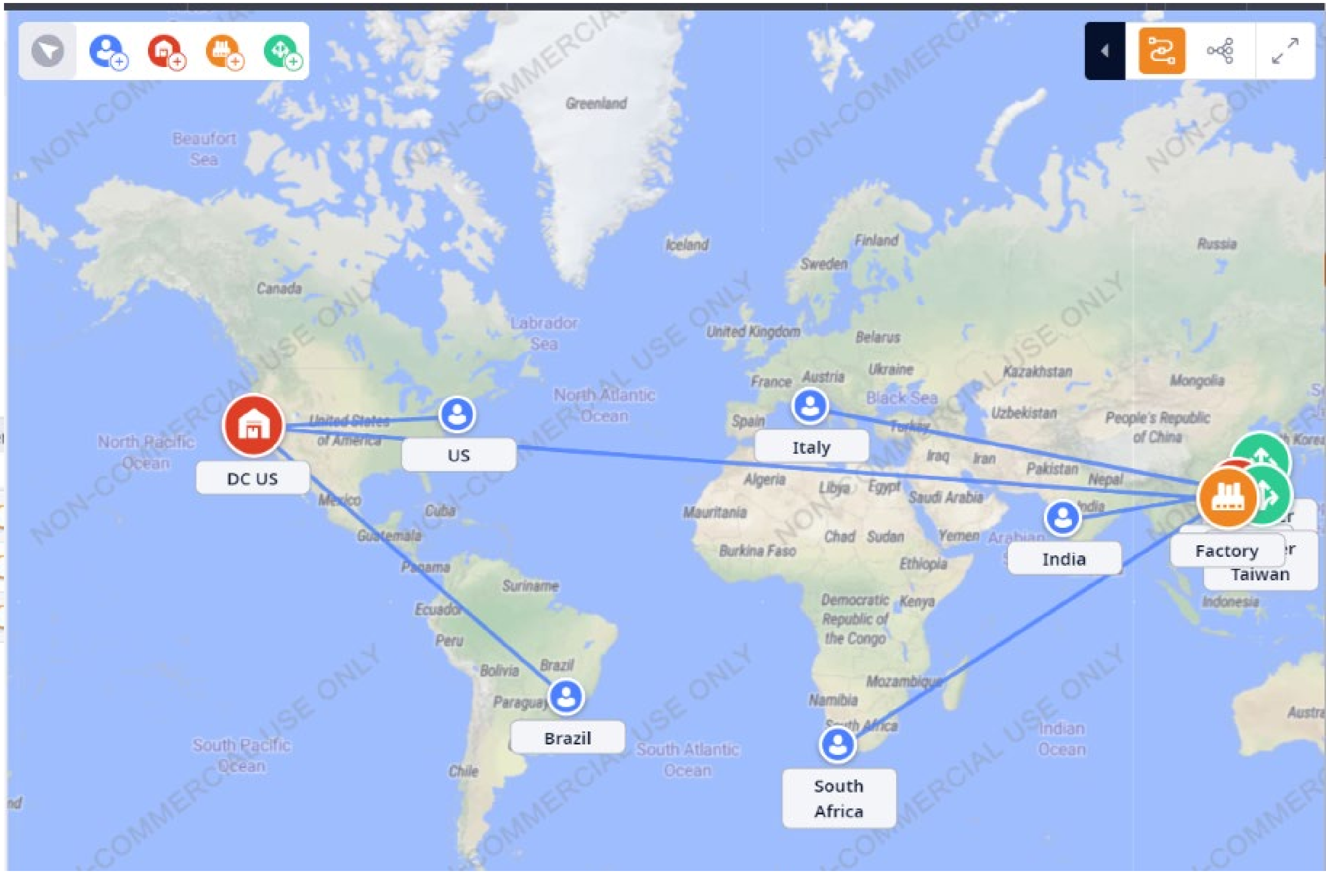


Co-funded by the European Union

Universidad de Alcalá

Multiple Suppliers (Dual-Sourcing)

- We changed from a single-source policy to a multiple-source policy for deliveries from distribution centers to customers.



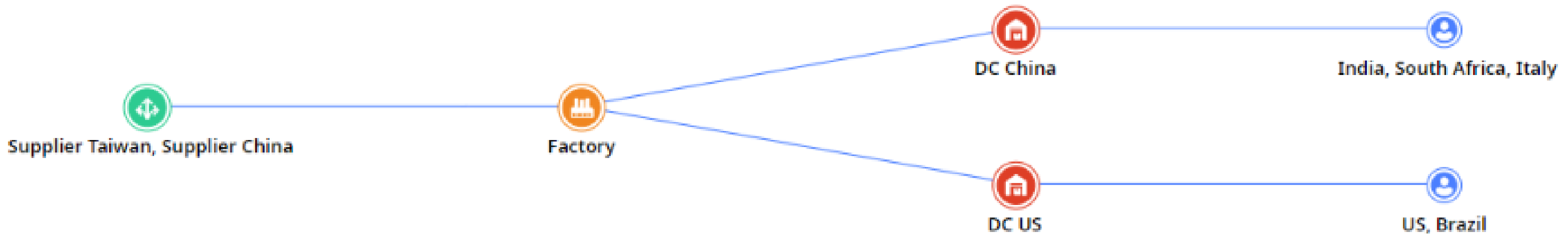
filter

Customers (5)

DCs and Factories (3)

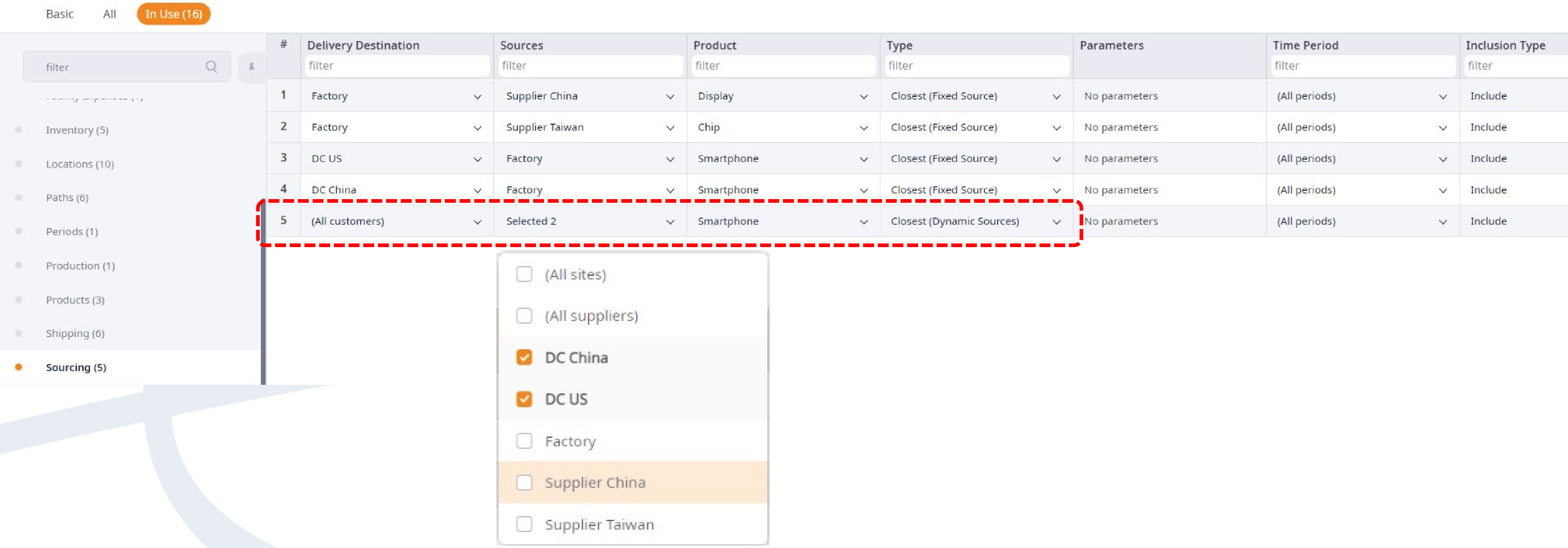
#	Name	Type	Location	Initially Oper
	filter	filter	filter	filter
1	DC US	DC	DC US location	
2	Factory	Factory	Factory location	
3	DC China	DC	DC China location	

- The new distribution center in China must be created!



Multiple Suppliers (Dual-Sourcing)

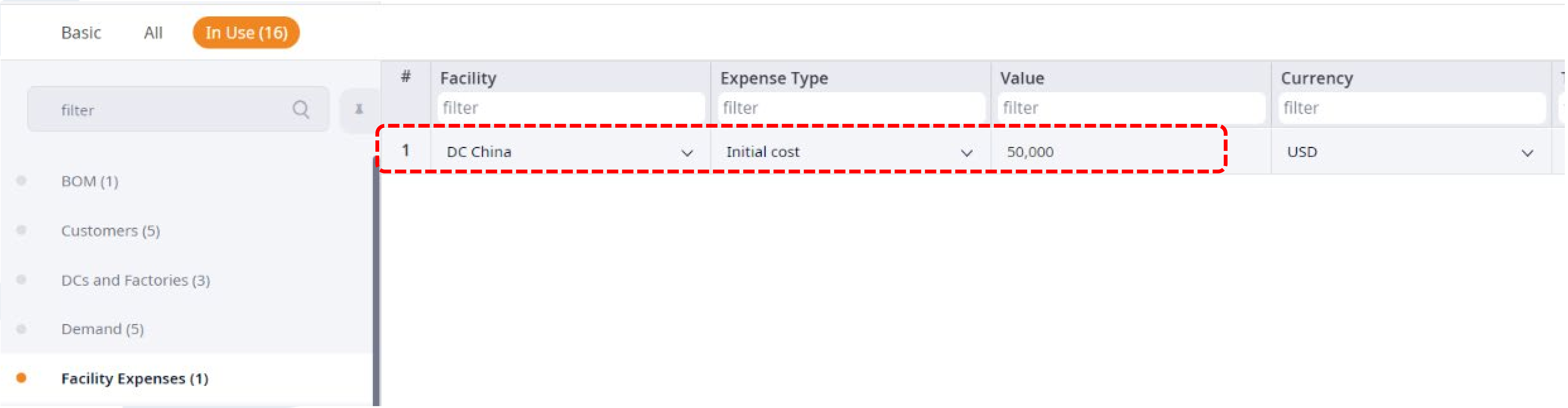
- We moved from the "Nearest Fixed Origin" sourcing policy to "Nearest Dynamic Origins".



#	Delivery Destination	Sources	Product	Type	Parameters	Time Period	Inclusion Type
1	Factory	Supplier China	Display	Closest (Fixed Source)	No parameters	(All periods)	Include
2	Factory	Supplier Taiwan	Chip	Closest (Fixed Source)	No parameters	(All periods)	Include
3	DC US	Factory	Smartphone	Closest (Fixed Source)	No parameters	(All periods)	Include
4	DC China	Factory	Smartphone	Closest (Fixed Source)	No parameters	(All periods)	Include
5	(All customers)	Selected 2	Smartphone	Closest (Dynamic Sources)	No parameters	(All periods)	Include

Multiple Suppliers (Dual-Sourcing)

- The opening of a new center will increase our fixed costs to $50,000.



Multiple Suppliers (Dual-Sourcing)

- It is also necessary to open the routes to the new center…

filter

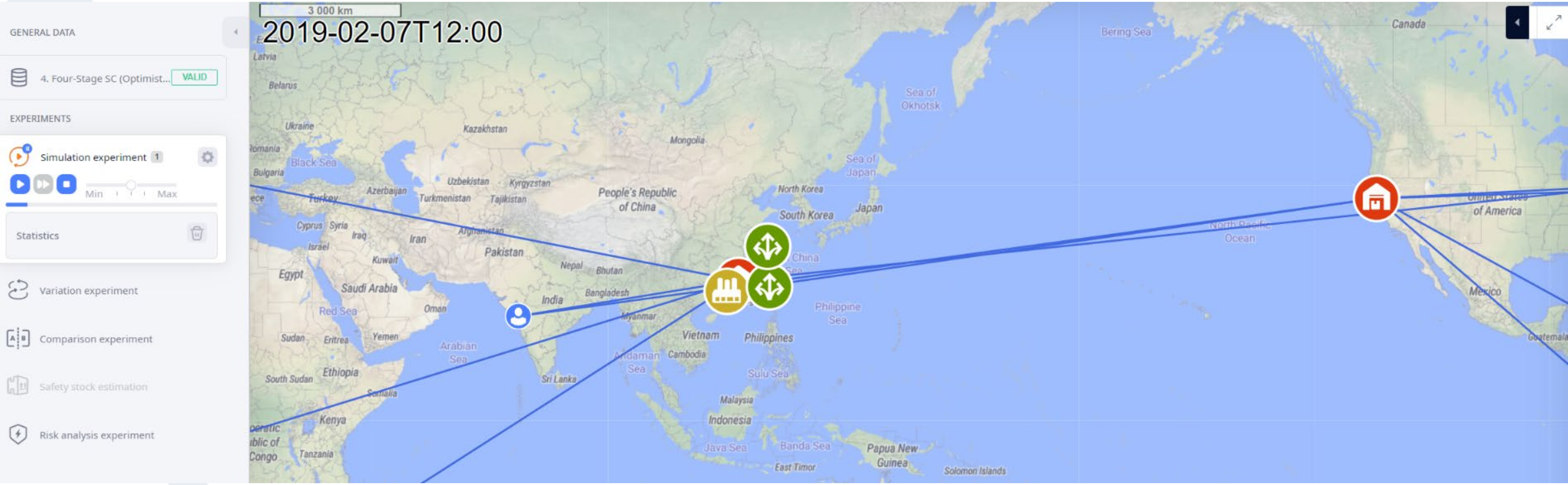
- BOM (1)
- Customers (5)
- DCs and Factories (3)
- Demand (5)
- Facility Expenses (1)
- Inventory (5)
- Locations (10)
- **Paths (6)**

#	From	To	Cost Calculation	Cost Calculation Parar	CO2 Calculation Parar	Currency	Distance	Distance Unit	Transportation Time	Time Unit
	filter	filter	filter			filter	filter	filter		filter
1	Supplier China locat	Factory location	Distance-based witl	0.5 * distance + 0 pe...	0 * distance + 0 per ...	USD	0	km	0	day
2	Supplier Taiwan loca	Factory location	Distance-based witl	0.8 * distance + 0 pe...	0 * distance + 0 per ...	USD	0	km	0	day
3	Factory location	DC US location	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	2	day
4	DC US location	(All locations)	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	0	day
5	Factory location	DC China location	Product&distance-b	0.005 * product (m³) ...	0 * product (m³) * di...	USD	0	km	0	day
6	DC China location	(All locations)	Product&distance-b	0.005 * product (m³) ...	0 * product (m³) * di...	USD	0	km	0	day

Straight	Vehicle Type	Time Period	Name	Inclusion Type
filter	filter	filter	filter	filter
	Truck	(All periods)		Include
	Ferry	(All periods)		Include
	Airplane	(All periods)		Include
	Airplane	(All periods)		Include
	Truck	(All periods)		Include
	Airplane	(All periods)		Include

Simulation

- We also see intercontinental freight...



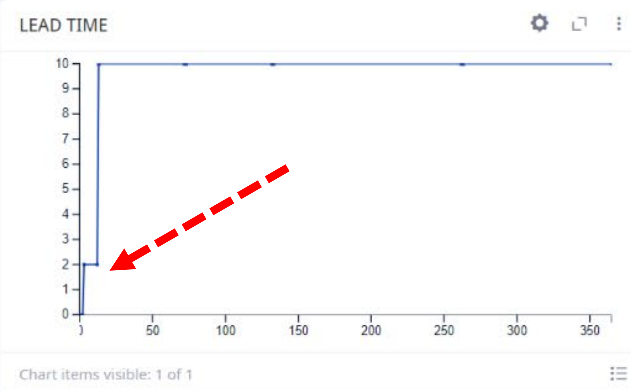
Run the simulation and compare

- **Financial** efficiency before:

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
1	Profit	1,985,485.255	USD
2	Production Cost	194,750	USD
3	Revenue	2,181,000	USD
4	Total Cost	195,514.745	USD
5	Transportation Cost	764.745	USD



- **Financial** efficiency after the change:

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
1	Profit	2,020,370.5	USD
2	Production Cost	190,250	USD
3	Revenue	2,211,000	USD
4	Total Cost	190,629.5	USD
5	Transportation Cost	379.5	USD

Run the simulation and compare

- **Operational** efficiency before:

- **Operational** efficiency after the change:

Run the simulation and compare

- Efficient **production and supply** before:

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	194,750	USD
2	Transportation Cost	DC	349.55	USD
3	Transportation Cost	Factory	415.196	USD

DEMAND (ORDERS BACKLOG), PRODUCTS PRODUCED, FULF...

#	Statistics	Value	Unit
1	Demand (Orders Back...	0	Order
2	Demand Placed (Drop...	4	Order
3	Demand Placed (Orde...	185	Order
4	Fulfillment Received (...	181	Order
5	Fulfillment Received (...	181	Order
6	Products Produced	3,895	pcs

- **Production and supply** efficiency after the changeover: :

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	190,250	USD
2	Transportation Cost	DC China	63.472	USD
3	Transportation Cost	DC US	110.44	USD
4	Transportation Cost	Factory	205.588	USD

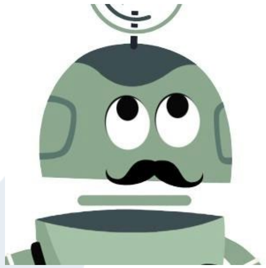
DEMAND (ORDERS BACKLOG), PRODUCTS PRODUCED, FULF...

#	Statistics	Value	Unit
1	Demand Placed (Drop...	1	Order
2	Demand Placed (Orde...	185	Order
3	Fulfillment Received (...	184	Order
4	Fulfillment Received (...	184	Order
5	Products Produced	3,805	pcs

Scenario comparison

KPI	Optimistic Scenario AS-IS Supply Chain Design	Optimistic Scenario Supply Chain Redesign “single distribution center - increased capacity”	Optimistic Scenario Supply Chain Redesign “new distribution center – dual sourcing”
Financial and customer performance:			
Production cost, $	93 250.0	194 000.0	190 250.0
Profit, $	1 007 437.706	1 985 485.255	2 020 370.5
Revenue, $	1 101 000.0	2 181 000.0	2 221 000.0
Total cost, $	93 562.294	195 514.745	190 629.5
Transportation cost (distribution center US), $	110.44	764.745	110.44
Transportation cost (distribution center China), $	-	-	63.472
Transportation cost (Factory), $	201.854	411.37	205.558
Service level, %	100%	100%	100%
Lead time, days	4	10	2.1
Operational performance:			
Maximum capacity usage in the supply chain, pcs	50	200	170
Maximum inventory in the supply chain (distribution center US), pcs	50	200	50
Maximum inventory in the supply chain (distribution center China), pcs	-	-	120
Maximum inventory in the supply chain (Factory), pcs	60	240	240
Production and sourcing performance:	0	0	
Current backlog orders	0	0	0
Customer delayed orders	0	0	0
Customer dropped orders	112.0	0	1.0
Customer in-time orders	73.0	185.0	184.0
Customer orders	185.0	185.0	185.0
Customer orders arrived	73.0	185.0	184.0
Produced, pcs	1865.0	3 895.0	3 805.0

Scenario comparison



- To compare the models in the table and highlight the pros and cons of each option, as well as justify and analyze the reasons for the change, we can rely on the data provided on pages 120 and 121 of the document.
- AS-IS Supply Chain Design:
  - Pros:
    - Low transportation cost from the U.S. distribution center ($110.44) .
    - Maximum inventory capacity in the U.S. distribution center of 50 units.
  - Cons:
    - Higher production costs ($194,000) compared to redesign options .
    - Lower production capacity (1865 units) compared to the redesign options .
- Supply Chain Redesign "Single Distribution Center - Increased Capacity":
  - Pros:
    - Increased production capacity (3895 units) compared to the current design .
    - Higher profit (1,985,485.255) and lower total cost (195,514.745) compared to the current design.
  - Cons:
    - Increase in transportation costs from China ($201,854) .
    - Lower inventory capacity at the distribution center in China (120 units) compared to other options.
- Supply Chain Redesign "New Distribution Center - Dual Sourcing":
  - Pros:
    - Higher profit (2,020,370.5) and lower total cost (190,629.5) compared to the other options.
    - Increased inventory capacity at the distribution center in China (170 units) compared to other options.
  - Cons:
    - Higher transportation costs from the factory ($205,558) compared to other options.
    - Increased transportation costs from China ($201,854) compared to the current design .
- Justification and Analysis of Change:
  - The decision to change the design of the supply chain is based on the search for improvements in key performance indicators such as profit, total costs and production capacity.
  - The introduction of a new dual-supply distribution center can result in higher profits and lower total costs compared to the current design and the option of a single distribution center with increased capacity.
  - However, it is important to consider the additional transportation costs associated with this option and evaluate whether the benefits outweigh these additional costs.
  - In summary, when comparing the models and their pros and cons, the supply chain redesign option with a new distribution center and dual supply appears to be the most favorable in terms of total benefits and costs, although the additional transportation costs involved must be carefully considered.

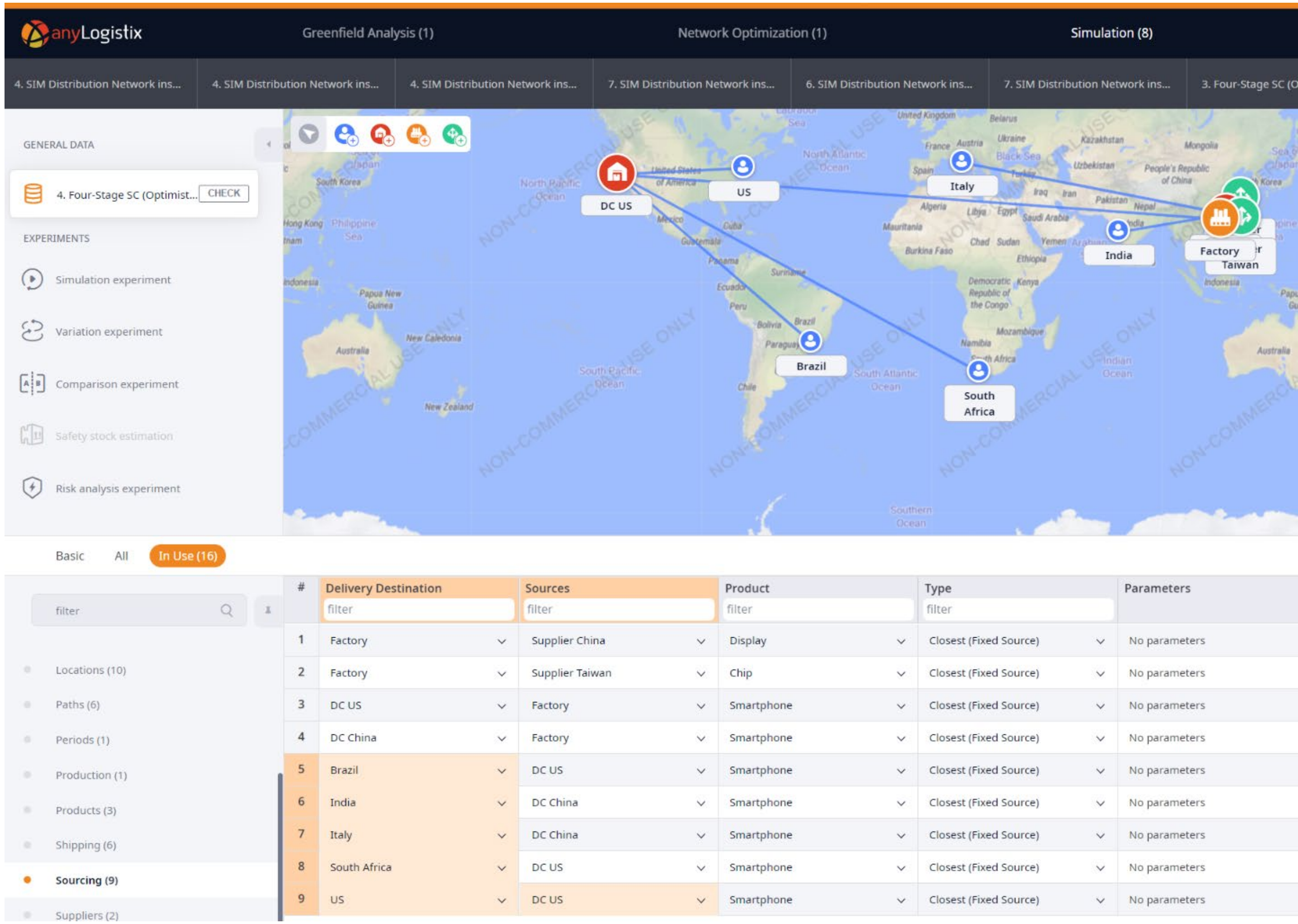
Analysis

- the redesigned supply chain performs much better than the original supply chain design (AS-IS) and the first supply chain redesign option.
- Financial, customer and operational performance has improved, and the WHC can double its total profits compared to the first supply chain redesign option.
- The results also evidence the maximum distribution center capacity needed for the new China distribution center (170 m3), as well as the production capacity (3,605 units).
- A more detailed analysis should include warehousing costs for the second distribution center in China.

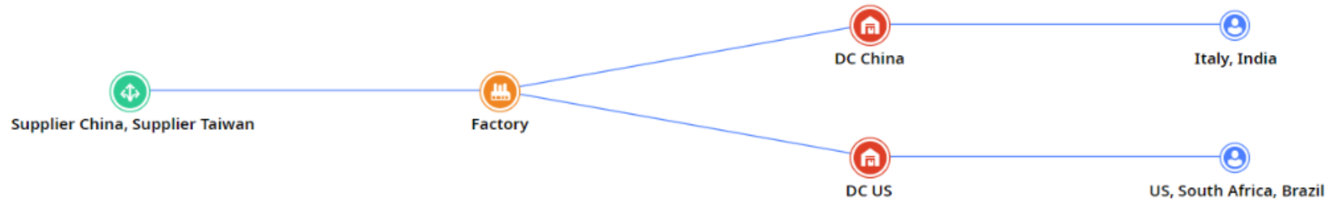
Scenario 5: Static Dual-Sourcing

Static Dual-Sourcing

- To estimate whether a dual sourcing policy will perform better than a single sourcing policy, we simulate the same example but with a fixed supplier policy.
  a) The U.S. distribution center delivers to customers in the U.S. and Brazil.
  b) China distribution center delivers to all other customers



#	Delivery Destination	Sources	Product	Type	Parameters
1	Factory	Supplier China	Display	Closest (Fixed Source)	No parameters
2	Factory	Supplier Taiwan	Chip	Closest (Fixed Source)	No parameters
3	DC US	Factory	Smartphone	Closest (Fixed Source)	No parameters
4	DC China	Factory	Smartphone	Closest (Fixed Source)	No parameters
5	Brazil	DC US	Smartphone	Closest (Fixed Source)	No parameters
6	India	DC China	Smartphone	Closest (Fixed Source)	No parameters
7	Italy	DC China	Smartphone	Closest (Fixed Source)	No parameters
8	South Africa	DC US	Smartphone	Closest (Fixed Source)	No parameters
9	US	DC US	Smartphone	Closest (Fixed Source)	No parameters



Run the simulation and compare

- **Financial** efficiency before:

Financial and customer performance
Operational performance
Production and Sourcing
Add Page

TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
1	Profit	2,020,370.5	USD
2	Production Cost	190,250	USD
3	Revenue	2,211,000	USD
4	Total Cost	190,629.5	USD
5	Transportation Cost	379.5	USD

- **Financial** efficiency after the change:

Financial and customer performance
Operational performance
Production and Sourcing
Add Page

TRANSPORTATION COST, REVENUE, PRODUCTION COST, TO...

#	Statistics	Value	Unit
1	Profit	1,816,893.662	USD
2	Production Cost	171,750	USD
3	Revenue	1,989,000	USD
4	Total Cost	172,106.338	USD
5	Transportation Cost	356.338	USD

Run the simulation and compare

- Operational efficiency before:

- Operational efficiency after the change:

Run the simulation and compare

- Efficient **production and supply** before:

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	190,250	USD
2	Transportation Cost	DC China	63.472	USD
3	Transportation Cost	DC US	110.44	USD
4	Transportation Cost	Factory	205.588	USD

DEMAND (ORDERS BACKLOG), PRODUCTS PRODUCED, FULF...

#	Statistics	Value	Unit
1	Demand Placed (Drop...	1	Order
2	Demand Placed (Orde...	185	Order
3	Fulfillment Received (...	184	Order
4	Fulfillment Received (...	184	Order
5	Products Produced	3,805	pcs

- **Production and supply** efficiency after the changeover: :

Financial and customer performance
Operational performance
Production and Sourcing
+ Add Page

TRANSPORTATION COST, PRODUCTION COST

#	Statistics	Object	Value	Unit
1	Production Cost	Factory	171,750	USD
2	Transportation Cost	DC China	41.049	USD
3	Transportation Cost	DC US	110.44	USD
4	Transportation Cost	Factory	204.849	USD

DEMAND (ORDERS BACKLOG), PRODUCTS PRODUCED, FULF...

#	Statistics	Value	Unit
1	Demand Placed (Drop...	38	Order
2	Demand Placed (Orde...	185	Order
3	Fulfillment Received ...	147	Order
4	Fulfillment Received ...	147	Order
5	Products Produced	3,435	pcs

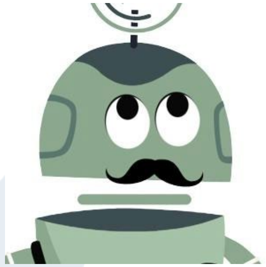
Scenario comparison

EUGLOH

KPI	Optimistic Scenario Supply Chain Redesign "single distribution center - increased capacity"	Optimistic Scenario Supply Chain Redesign "new distribution center – dual sourcing"	Optimistic Scenario Supply Chain Redesign "new distribution center – single sourcing"
Financial and customer performance:			
Production cost, $	194 000.0	190 250.0	190 250.0
Profit, $	1 985 485.255	2 020 370.5	2 020 370.5
Revenue, $	2 181 000.0	2 221 000.0	2 221 000.0
Total cost, $	195 514.745	190 629.5	190 629.5
Transportation cost (distribution center US), $	764.745	110.44	110.44
Transportation cost (distribution center China), $	-	63.472	63.472
Transportation cost (Factory), $	411.37	205.558	205.558
Service level, %	100%	100%	100%
Lead time, days	10	2.1	2.1
Operational performance:			

Operational performance:			
Maximum capacity usage in the supply chain, pcs	200	170	170
Maximum inventory in the supply chain (distribution center US), pcs	200	50	50
Maximum inventory in the supply chain (distribution center China), pcs	-	120	120
Maximum inventory in the supply chain (Factory), pcs	240	240	240
Production and sourcing performance:			
Current backlog orders	0	0	0
Customer delayed orders	0	0	0
Customer dropped orders	4.0	1.0	1.0
Customer in-time orders	181.0	184.0	184.0
Customer orders	185.0	185.0	185.0
Customer orders arrived	181.0	184.0	184.0
Produced, pcs	3 895.0	3 805.0	3 805.0

Scenario comparison



- To compare the models in the table and highlight the pros and cons of each option, as well as justify and analyze the reasons for the change, we can consider the following points:

- ## Single Distribution Center - Increased Capacity:
  - Pros:
    - Increased production capacity.
    - Lower transportation costs compared to a new distribution center scenario.
    - Maximum capacity utilization in the supply chain.
  - Cons:
    - Higher lead time compared to the other scenarios.
    - Increased inventory in the U.S. distribution center.
    - Justification for the change:
  - Despite higher production capacity and lower transportation costs, the higher lead time and higher inventory in the U.S. distribution center could affect the overall efficiency of the supply chain.

- ## New Distribution Center - Dual Sourcing:
  - Pros:
    - Lower lead time compared to a single distribution center scenario.
    - Reduced inventory at distribution centers.
    - Higher profitability compared to the other scenarios.
  - Cons:
    - Higher transportation costs compared to a single distribution center scenario.
  - Justification for the change:
    - The significant reduction in lead time and inventory, together with the increase in profit, could justify the implementation of this scenario despite the higher transportation costs.

- ## New Distribution Center - Single Sourcing:
  - Pros:
    - Lower lead time compared to a single distribution center scenario.
    - Reduced inventory at distribution centers.
    - Lower transportation costs compared to the dual-supply scenario.
  - Cons:
    - Lower profit compared to the dual supply scenario.
  - Justification for the change:
    - Although it is less profitable compared to the dual sourcing scenario, the benefits of lower lead time and lower transportation costs may be appropriate depending on the company's priorities.

- ## In summary,
  - the choice between the different supply chain redesign scenarios will depend on the company's specific objectives, such as cost optimization, delivery time improvement or profit maximization. Each option has its advantages and disadvantages, and the final decision should be based on a detailed analysis of how each scenario affects business and operational results.

So that's it...

Ni-Hao!